

SAT UZBEKISTAN

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Module-1

1.

A group of 40 employees selected at random from all the employees at a company were surveyed about the number of books they read last year. From the data collected, it was estimated that the mean number of books employees at the company read last year is 6.8, with an associated margin of error of 1.1. Which of the following is the most appropriate conclusion?

- (A) It is plausible that the actual mean number of books employees at the company read last year is less than 5.7.
- (B) It is plausible that the actual mean number of books employees at the company read last year is between 5.7 and 7.9.
- (C) It is plausible that every employee at the company read between 6.8 and 10.1 books last year.
- (D) It is plausible that the actual mean number of books employees at the company read last year is greater than 7.9.

2.

In triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . In triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?

- (A) The measure of angle F
- (B) The measure of angle T
- (C) The measure of angle F and the measure of angle T
- (D) No additional information is needed.

3.

In the xy -plane, the graph of function f , where $y = f(x)$, has exactly 8 x -intercepts. One of these x -intercepts is $(17, 0)$. The rational function g is defined by $g(x) = \frac{f(x)}{x-17}$. In the xy -plane, how many x -intercepts does the graph of $y = g(x)$ have?

4. While the mass of an object is the same everywhere, the weight of an object is not the same on different planets. An object has a weight of 170.00 pounds on Earth and a weight of 186.32 pounds on Neptune. The object's weight on Jupiter is 252.8% of its weight on Earth. If the object's weight on Neptune is $x\%$ of its weight on Jupiter, which of the following is closest to the value of x ?

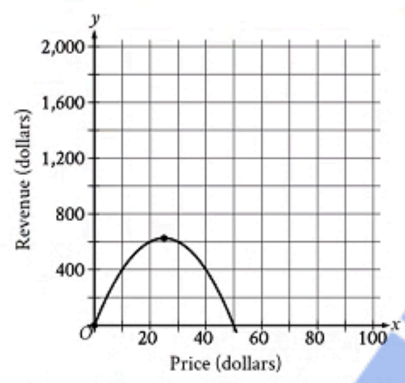
- (A) 43.35
- (B) 71.73
- (C) 277.07
- (D) 429.76

5.

$$\begin{aligned} -x - wy &= -357 \\ 2x - wy &= 51 \end{aligned}$$

In the given system of equations, w is a constant. In the xy -plane, the graphs of these equations intersect at the point $(q, 17)$, where q is a constant. What is the value of w ?

6.



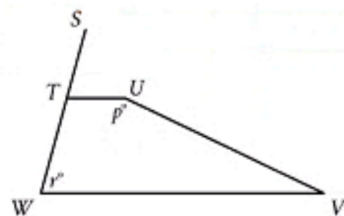
The graph gives the estimated revenue, in dollars, for a company when the price of the company's product is x dollars. What does the y -value of the vertex of the graph represent in this context?

- (A) The price, in dollars, of the company's product
- (B) The minimum estimated revenue, in dollars
- (C) The maximum estimated revenue, in dollars
- (D) The estimated revenue, in dollars, when the price of the company's product is 0 dollars

7.

A linear model estimates the population of a city from 1993 to 2017. The model estimates the population was 52 thousand in 1993, 219 thousand in 2013, and x thousand in 2017. To the nearest whole number, what is the value of x ?

8.



Note: Figure not drawn to scale.

In the figure shown, point T lies on segment SW , and segment TU is parallel to segment VW . The measure of angle V is 26° , and the measure of angle STU is 74° . What is the value of $p - r$?



9.

On January 1, there were 233 views of a video posted on a social media site. Every 2 days after January 1, the number of views of the video had increased by 60% of the number of views 2 days earlier. The function f gives the predicted number of views x days after January 1. Which equation defines f ?

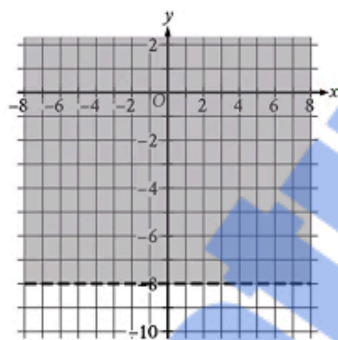
(A) $f(x) = 233(0.60)^{\frac{x}{2}}$

(B) $f(x) = 233(0.60)^{2x}$

(C) $f(x) = 233(1.60)^{\frac{x}{2}}$

(D) $f(x) = 233(1.60)^{2x}$

10.



The shaded region shown represents the solutions to the inequality $ry < 128$, where r is a constant. What is the value of r ?

(A) 16

(B) 8

(C) -8

(D) -16

11.

$$f(x) = 3,950(0.38)^{\frac{x}{12}}$$

The function f gives the value, in dollars, of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what is the value of p ?

Ⓐ 3

Ⓑ 8

Ⓒ 38

Ⓓ 62

12.

A beekeeper's initial observation of the population of a certain bee colony was 1,100 bees. The beekeeper set a goal to increase the population to 2,600 bees. The beekeeper uses a model that predicts the population of this bee colony begins at 1,100 and increases by 120 bees per week in the first two weeks after the initial observation, and then increases by 180 bees per week until the beekeeper's goal is reached. According to this model, at the end of week w after the initial observation, where $w > 2$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

Ⓐ $p(w) = 2,600 - 180w$ Ⓑ $p(w) = 2,480 + 180w$ Ⓒ $p(w) = 1,620 - 180w$ Ⓓ $p(w) = -120 + 180w$

13.

Function f is a quadratic function. The graph of $y = f(x)$ in the xy -plane has a vertex at $(4, -6)$, contains the point $(3, -8)$, and has a y -intercept at $(0, a)$. The graph of $y = 5 \cdot f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

14.

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week, the number of prey had increased by 2,700% of the number of prey at the start of the week, and the number of predators had increased by 180% of the number of predators at the start of the week. The number of prey at the end of the week was $p\%$ greater than the number of predators at the end of the week. What is the value of p ?

15.

Sonja purchased a laptop in September 2018 when the laptop had a value of \$1,000. Each month for the next 50 months, the value of the laptop decreased by 2% of the previous month's value. Which equation represents this situation, where v is the value of the laptop, in dollars, m is the number of months since Sonja purchased the laptop, and $m \leq 50$?

(A) $v = 2,018(0.98)^m$

(B) $v = 2,018(2)^m$

(C) $v = 1,000(0.98)^m$

(D) $v = 1,000(2)^m$

16.

In triangle FGH , the measure of angle F is 63° . Which additional piece of information is sufficient to prove that triangle FGH is a right triangle?

(A) The length of $\overline{FH}$ is 63.

(B) The length of $\overline{GH}$ is 126.

(C) The measure of angle H is 27° .

(D) The sum of the measures of angles F , G , and H is 180° .

17.

A survey of the daily commuting times, in minutes, of seven different Chicago-based employees at a certain company yielded a data set consisting of the values 31, 13, 44, 32, 53, 22, and 44. A survey of the daily commuting times, in minutes, of seven different San Francisco-based employees at the same company yielded a data set consisting of the values 46, 31, 19, 33, 48, x , and 38. If the means of the two data sets are equal, what is the value of x ?

18.

A survey was conducted in Madison to estimate the percent of adult residents who support the addition of streetlights at every corner in their town. Based on the survey, it is estimated that 40% of all adult residents of Madison support the addition of streetlights at every corner in their town, with an associated margin of error of 6%. Which of the following is the most appropriate conclusion about all adult residents of Madison?

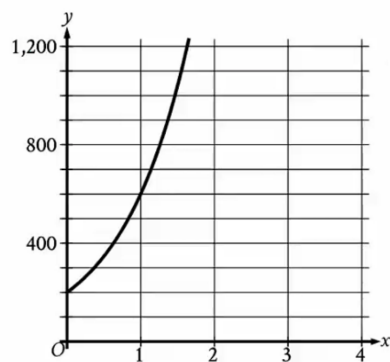
(A) Plausible values for the percent of all adult residents of Madison who support the addition of streetlights at every corner in their town are between 34% and 46%.

(B) The addition of streetlights at every corner in the town is supported by exactly 34% of all adult residents of Madison.

(C) Plausible values for the percent of all adult residents of Madison who support the addition of streetlights at every corner in their town are either less than 34% or more than 46%.

(D) The addition of streetlights at every corner in the town is supported by more than 46% of all adult residents of Madison.

19.



The graph models the number of online newsletter subscribers at the end of every six-month period, where x is the number of six-month periods since the end of January 1995 and $0 \leq x \leq 4$. Which statement is the best interpretation of the point $(1, 600)$ in this context?

- (A) The estimated number of online newsletter subscribers at the end of the first six-month period was 600.
- (B) The estimated number of online newsletter subscribers increased every six months by 600 subscribers.
- (C) The estimated number of online newsletter subscribers at the end of January 1995 was 600.
- (D) The estimated number of online newsletter subscribers at the end of January 1996 was 600.

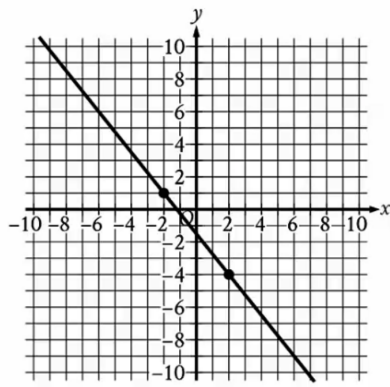
20.

The table summarizes the number of objects in each group.

Group	Number of objects
A	325
B	64
C	611
D	96
Total	1,096

The number of objects in group C is $p\%$ of the number of objects in group A. What is the value of p ?

21.



Line k is shown in the xy -plane. Line j (not shown) is perpendicular to line k and passes through the point $(7, 15)$. What is the y -coordinate of the y -intercept of line j ?

22.

A baker has a total of 15 cups of flour and enough other ingredients to make banana pancakes and apple pancakes. The baker needs 1.25 cups of flour for a batch of banana pancakes and 0.75 cups of flour for a batch of apple pancakes. If the baker makes 8 batches of banana pancakes, what is the greatest number of whole batches of apple pancakes the baker can make?

(A) 6

(B) 7

(C) 8

(D) 9

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Module-2

1.

$$x^2 - 11x - 3 = 0$$

What is a solution to the given equation?

(A) $\frac{11 + \sqrt{(-11)^2 - 4(1)(-3)}}{2(-3)}$

(B) $\frac{11 + \sqrt{(-11)^2 - 4(1)(-3)}}{2(1)}$

(C) $\frac{3 + \sqrt{(-3)^2 - 4(1)(-11)}}{2(-3)}$

(D) $\frac{3 + \sqrt{(-3)^2 - 4(1)(-11)}}{2(1)}$

2.

Ari examined a set of 86 plants. The lightest plant had a mass of 2.8 kilograms (kg) and the heaviest plant had a mass of 6.3 kg. Chihiro examined the same set of 86 plants and also an additional plant with a mass of 11.6 kg. Which of the following must be true about the set of plants that Ari examined and the set of plants that Chihiro examined?

(A) The standard deviation of the masses, in kg, of the plants that Ari examined is greater than the standard deviation of the masses, in kg, of the plants that Chihiro examined.

(B) The range of the masses, in kg, of the plants that Ari examined is greater than the range of the masses, in kg, of the plants that Chihiro examined.

(C) The median of the masses, in kg, of the plants that Ari examined is less than the median of the masses, in kg, of the plants that Chihiro examined.

(D) The mean of the masses, in kg, of the plants that Ari examined is less than the mean of the masses, in kg, of the plants that Chihiro examined.

3.

A machine makes 8-inch, 5-inch, and 7-inch parts. During a certain day, the number of 8-inch parts that the machine makes is 3 times the number n of 5-inch parts, and the number of 7-inch parts is 12. During this day, the machine makes 100 parts total. Which equation represents this situation?

(A) $8(3n) + 5n + 7(12) = 100$

(B) $8n + 5n + 7n = 100$

(C) $3n + 12 = 100$

(D) $4n + 12 = 100$

4.

An object's speed, in meters per hour, is increasing at a rate of 1,490 meters per hour per second. Which of the following is closest to this rate in **feet per second squared**? (Use 1 meter = 3.28 feet.)

(A) 0.126

(B) 1.36

(C) 7.57

(D) 81.5

5.

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(32) > 0$, $f(35) < 0$, and $f(38) > 0$, which of the following could be the value of $a + b$?

(A) 35

(B) 37

(C) 67

(D) 69

6.

One gallon of sealant costs \$13 and will cover 300 square feet of a surface. A deck has a total surface area of d square feet. Which equation represents the cost c , in dollars, of the sealant needed to cover the deck twice?

(A) $c = \frac{300d}{13}$

(B) $c = \frac{600d}{13}$

(C) $c = 13\left(\frac{d}{150}\right)$

(D) $c = 13\left(\frac{d}{300}\right)$

7.

$$26z^{18} + bz^9 + 70$$

In the given expression, b is a positive integer. If $qz^9 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b ?

8.

In isosceles triangle ABC , sides AB and AC are congruent. Point D divides side BC such that the length of $\overline{BD}$ is $\frac{2}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD . If the length of $\overline{BE}$ is 14, what is the length of $\overline{CF}$?

☐ (A) 28

☐ (B) 35

☐ (C) 49

☐ (D) 98

9.

In the xy -plane, a unit circle with center at the origin O contains point A with coordinates $(1, 0)$ and point B with coordinates $\left(-\frac{5}{\sqrt{89}}, \frac{8}{\sqrt{89}}\right)$. If the measure of angle AOB is w radians, what is the value of $\tan w$?

10.

In triangle ABC and triangle DEF , sides AB and DE each have a side length of 10 inches, and angles A and D each have an angle measure of 40° . Which of the following additional pieces of information is(are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?

- I. The measures of angles B and C are equal.
- II. The lengths of sides AC and DF are equal.
- III. The lengths of sides BC and EF are equal.

☐ (A) I is sufficient, but II and III are not.

☐ (B) II is sufficient, but I and III are not.

☐ (C) III is sufficient, but I and II are not.

☐ (D) II is sufficient and III is sufficient, but I is not.

11.

The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$ and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

- I. $f(x) = a(4.2)^{x+b}$
- II. $g(x) = a(4.2)^x + b$

☐ (A) I only

☐ (B) II only

☐ (C) I and II

☐ (D) Neither I nor II

12.

A circle in the xy -plane has its center at $(-7, 3)$ and has a radius of 9. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

13.

One gallon of sealant costs \$29 and will cover 300 square feet of a surface. A deck has a total surface area of d square feet. Which equation represents the cost c , in dollars, of the sealant needed to cover the deck twice?

(A) $c = \frac{300d}{29}$

(B) $c = \frac{600d}{29}$

(C) $c = 29\left(\frac{d}{150}\right)$

(D) $c = 29\left(\frac{d}{300}\right)$

14.

The function f is defined by $f(x) = -39^x$. The function g is a decreasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points, (h, j) and (k, m) , where $j > m$. When $g(x) < f(x)$, which of the following must also be true?

(A) $x > k$

(B) $x < h$

(C) $x > k$ or $x < h$

(D) $h < x < k$

15.

A conservation specialist hung artificial nesting structures each in the shape of a right rectangular prism for a species of native owl. Each structure has a height of 11 inches. The length of each structure's base is x inches, which is 1 inch more than the width of the structure's base. Which function V gives the volume of each structure, in cubic inches, in terms of the length of the structure's base?

(A) $V(x) = 11x(x - 1)$

(B) $V(x) = 11x(x + 1)$

(C) $V(x) = x(x + 11)(x - 1)$

(D) $V(x) = x(x + 11)(x + 1)$

16.

For the linear function $p, p(c) = -2$, where c is a constant, $p(5) = 34$, and the slope of the graph of $y = p(x)$ in the xy -plane is 6. For the linear function $t, t(c) = -4$ and $t(6) = 52$. What is the slope of the graph of $y = t(x)$ in the xy -plane?

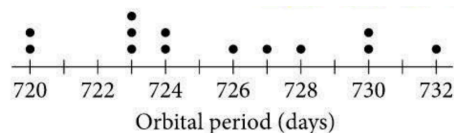
(A) -1

(B) 4

(C) 6

(D) 8

17.



A data set of the orbital periods, rounded to the nearest whole number of Earth days, for 13 of Jupiter's moons is represented in the dot plot. An additional moon with an orbital period of 251 days is added to the original data set to create a new data set of 14 orbital periods. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

(A) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

(B) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is less than the median of the original data set.

(C) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.

(D) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

18.

$$P(t) = 260(1.03)^{\left(\frac{1}{8}\right)t}$$

The function P models the population, in thousands, of a certain city t years after 2009. According to the model, the population is predicted to increase by $n\%$ every 8 months. What is the value of n ?

(A) 0.22

(B) 1.03

(C) 2

(D) 3

19.

At what value of x does the graph of $y = x^2 + 18x - 23$ reach its minimum in the xy -plane?

(A) -23

(B) -9

(C) 9

(D) 18

20.

James purchased a certain baseball card on January 1. The function $f(x) = 55(1.04)^x$, where $0 \leq x \leq 10$, gives the predicted value, in dollars, of the baseball card x years after James purchased it. What is the best interpretation of the statement " $f(7)$ is approximately equal to 72" in this context?

- (A) When the baseball card's predicted value is approximately 72 dollars, it is 7% greater than the predicted value, in dollars, on January 1 of the previous year.
- (B) When the baseball card's predicted value is approximately 72 dollars, it is 7 times the predicted value, in dollars, on January 1 of the previous year.
- (C) From the day James purchased the baseball card to 7 years after James purchased the card, its predicted value increased by a total of approximately 72 dollars.
- (D) 7 years after James purchased the baseball card, its predicted value is approximately 72 dollars.

21.

A circle in the xy -plane has its center at $(2, 9)$. Line t is tangent to this circle at the point $(a, -4)$, where a is a constant. The slope of line t is $\frac{6}{5}$. What is the value of a ?

- (A) $-\frac{68}{5}$
- (B) $-\frac{53}{6}$
- (C) $\frac{77}{6}$
- (D) $\frac{88}{5}$

22.

x	y
20	-20
24	12
28	-20

The table shows three values of x and their corresponding values of y , where $y = f(x) + 6$ and f is a quadratic function. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?